

EDUCATION

- Pamantasan ng Lungsod ng Valenzuela

Bachelor of Science in Information Technology

Valenzuela City, Metro Manila

2023 – 2027

EXPERIENCE

- Upturn Business Solutions PH

Software Engineer Intern

Valenzuela City, Metro Manila

Jan 2026 – Apr 2026

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Tax Records Portal:

Sole-engineered a production enterprise platform (React 19, Spring Boot 4, PostgreSQL) used by 80–100 employees managing 100+ clients, replacing manual tax compliance workflows company-wide; handled post-launch schema migrations and live data patches on a running production system with no data loss

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Access Control:

Two-layer enforcement: method-level @PreAuthorize backed by JWT-embedded permissions (no per-request DB lookups) and data-level query scoping per role, with 36 permissions across 7 roles and Bucket4j rate limiting

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Domain Architecture:

Multi-stage client lifecycle with lazy-created profile update drafts, partial unique index enforcing at-most-one active update per client, and server-computed JSONB diffs on approval; bulk task import with 500-row batches, partial success, and duplicate detection

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Operations & Infrastructure:

Analytics with role-differentiated query scoping, LEAD() window functions for rejection turnaround, and intentionally divergent rejection-cycle formulas between system-wide and per-accountant views; deployed on Railway/Cloudflare with Sentry, async emails via Resend, and optimistic locking

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Pamantasan ng Lungsod ng Valenzuela

Tech Lead, College CMS (8-person team)

Valenzuela City, Metro Manila

2025

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Leadership:

Led architecture decisions and full-stack integration (Next.js, FastAPI, PostgreSQL, Cloudinary) for a CMS commissioned by the College of Engineering and IT; owned the frontend-backend integration layer, drove system design, and managed team coordination across the full project lifecycle
- PROJECTS
- AGOS – Real-Time Waterway Monitoring System

IoT-integrated monitoring system built with FastAPI, React, PostgreSQL, and WebSocket (backend, admin dashboard, responder PWA)

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Real-Time Pipeline:

Ingests IoT sensor readings and camera frames, runs ML blockage detection, computes fusion risk scores, and broadcasts live updates to all connected clients

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Notification & Alerting:

Web Push with VAPID, SMS OTP via Android gateway, automated escalation for unacknowledged alerts, offline acknowledgement queue with auto-flush on reconnect

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Survey Management with Data Analysis

Enterprise survey platform built with React, TypeScript, ASP.NET Core, and SQL Server, with backend optimizations for high-volume response handling

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Survey Builder:

Modular design system with 10+ question types, validation rules, and conditional logic

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Analytics & Data:

Multi-dimensional analytics engine with trend visualizations, JWT-based RBAC, QR code access, and PDF/Excel exports

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PLV Cloud

File-sharing platform built with Laravel, Livewire, PostgreSQL, and Docker for Pamantasan ng Lungsod ng Valenzuela

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Platform:

Hierarchical folder structure with unlimited nesting, Cloudinary CDN, granular sharing permissions, queue-based background jobs, and Docker production deployment with Nginx and CI/CD

TECHNICAL SKILLS

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Languages:

JavaScript, TypeScript, Python, Java, C#

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Databases:

PostgreSQL, MySQL, MongoDB, SQL Server

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Frameworks:

React, Next.js, Spring Boot, Node.js, Express.js, .NET, FastAPI, Laravel, Tailwind CSS

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Cloud & Tools:

Railway, Cloudflare Pages/R2, Vercel, Docker, Git, GitHub, Postman, Swagger

AWARDS

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ITLympics 2025 – Gold Medal

Python Programming, Pamantasan ng Lungsod ng Valenzuela

Apr 2025

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14th IT Skills Olympics – Bronze Medal

Python Programming, University of Makati

Nov 2025

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ITLympics 2024 – Gold Medal

Java Programming, Pamantasan ng Lungsod ng Valenzuela

Apr 2024